MIL-S-43894(GL)

19 August 1974
SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

SHELVING KITS, STORAGE AND DISPLAY, WOOD:

FOR

BOXES, SHIPPING, METAL (CONEX AND MILVAN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for CONEX and MILVAN storage and display shelving kits fabricated from wood (see 6.1).

1.2 Classification. The shelving shall consist of the following kits as specified (see 6.2).

Type I - CONEX Kits

Kit I - 4 assemblies
Kit II - 2 assemblies

Type II - MILVAN Kits

Kit III - 9 assemblies
Kit IV - 9 assemblies

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals

FSC 7125

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PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-650 - Crates, Wood, Open and Covered
PPP-S-760 - Strapping, Nonmetallic (and Connectors)
PPP-T-97 - Tape; Pressure Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116 - Preservation Packaging, Methods Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

US Army Natick Laboratories

5-13-2414 - Shelving Kit, Storage and Display, Wood: For Box, Metal, Shipping (CONEX); Kit I
5-13-2415 - Shelving Kits, Storage and Display, Wood, For Box, Metal Shipping (CONEX) Center Bin Assembly
5-13-2416 - Shelving, Kit, Storage and Display, Wood, For Box, Metal Shipping (CONEX) Bin Assembly "B"
5-13-2417 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (CONEX) Bin Assembly "A"
5-13-2418 - Shelving, Kit, Storage and Display, Wood, For Box, Metal Shipping (CONEX) Kit II
5-13-2419 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (CONEX) Bin Assembly "C"
5-13-2420 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (CONEX) Bin Assembly "D"
5-13-2086 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Kit III
5-13-2087 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "E"
5-13-2088 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "F"
5-13-2089 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "H"

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5-13-2090 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Kit IV
5-13-2091 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "G"
5-13-2092 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "I"
5-13-2093 - Shelving, Kit, Storage and Display, Wood, For Box, Metal, Shipping (MILVAN) Bin Assembly "J"

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

US Department of Commerce

PS 1 - Softwood Plywood, Construction and Industrial
PS 51 - Hardwood and Decorative Plywood

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.2 and 6.4).

3.2 Material. Material required shall conform to the following:

3.2.1 Plywood. Plywood shall be any grade of exterior type plywood or shall be interior type, C-D grade (Sheathing, Int.) with exterior glue conforming to PS 1 or exterior type, 3-4 veneers conforming to PS 51.

3.2.2 Lumber and boards. Lumber and boards shall be kiln dried, commercial No. 2 common or better, surfaced on four sides (S4S).

3.2.3 Nails.

3.2.3.1 Box nails. Box nails shall be commercial No. 6d cement coated steel, flat head, diamond point (approximately 315 nails per pound).

3.2.3.2 Common nails. Common nails shall be commercial No. 8d bright steel, flat head, diamond point (approximately 95 nails per pound).

3.3 Design and construction. The shelving shall conform to the design, details and dimensions shown on the applicable drawings referenced in 2.1.

3.3.1 Type I, kit I. Kit I shall consist of all the unassembled component parts of two center bin assemblies, one bin assembly A and one bin assembly B, as shown on Drawing 5-13-2414, with details for each piece in each assembly as shown on Drawings 5-13-2415, 5-13-2416, and 5-13-2417. In addition, the following components shall be furnished with each kit in the quantities indicated, for use in the erection, installation and blocking of the shelving in each CONEX unit:

a. Two lbs. of box nails conforming to 3.2.3.1
b. One lb. of common nails conforming to 3.2.3.2
c. Eight pieces of 1/2 by 34 by 93-1/2 inch plywood conforming to 3.2.1
d. Two pieces of 1 by 3 by 93-1/2 inch boards conforming to 3.2.2
e. Four pieces of 2 by 4 by 93-1/2 inch lumber conforming to 3.2.2
f. Three pieces of 1/4 by 11-1/2 by 70 inch plywood conforming to 3.2.1

3.3.2 Type I, kit II. Kit II shall consist of all the unassembled component parts of one bin assembly C and one bin assembly D as shown on Drawing 5-13-2418, with details for each piece in each assembly as shown on Drawings 5-13-2419 and

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5-13-2420. The following additional components shall be furnished with each kit in the quantities indicated, for use in the erection, installation and blocking of the shelving in each CONEX unit:

a. Two lbs. of box nails conforming to 3.2.3.1
b. One lb. of common nails conforming to 3.2.4
c. Four pieces of 1/2 by 34 by 93-1/2 inch plywood conforming to 3.2.1
d. Three pieces of 1 by 3 by 93-1/2 inch board conforming to 3.2.2
e. Three pieces of 2 by 4 by 93-1/2 inch lumber conforming to 3.2.2
f. One piece 1/4 by 20 by 70 inch plywood, conforming to 3.2.1
g. One piece 1/4 by 24 by 70 inch plywood, conforming to 3.2.1

3.3.3 Type II, kit III. Kit III shall consist of all the unassembled component parts of one bin assembly E, two bin assemblies F, four bin assemblies G and two bin assemblies H as shown on Drawings 5-13-2086 with details for each piece in each assembly as shown on Drawings 5-13-2087, 5-13-2088, 5-13-2089, and 5-13-2091. In addition, the following components shall be furnished with each kit in the quantities indicated, for use in the erection, installation and blocking of the shelving in each MILVAN.

a. Two pounds of box nails conforming to 3.2.3.1
b. Three pounds of common nails conforming to 3.2.3.2
c. One piece of 2" x 4" x 90" lumber conforming to 3.2.2
d. Four pieces of 2" x 4" x 78-1/2" lumber conforming to 3.2.2
e. Eleven pieces of 2" x 4" x 93-1/2" lumber conforming to 3.2.2
f. One piece of 1/2" x 24" x 81-1/2" plywood conforming to 3.2.1
g. One piece of 1/2" x 20" x 81-1/2" plywood conforming to 3.2.1
h. One piece of 1/2" x 11-3/4" x 81-1/2" plywood conforming to 3.2.1
i. Sixteen pieces of 1/2" x 40" x 93-1/2" plywood conforming to 3.2.1
j. Two pieces of 1/2" x 40" x 90" plywood conforming to 3.2.1

3.3.4 Type II, kit IV. Kit IV shall consist of all the unassembled component parts of one bin assembly E, two bin assemblies G, three bin assemblies I and three bin assemblies J as shown on Drawings 5-13-2090 with details for each piece in each assembly as shown on Drawings 5-13-2087, 5-13-2091, 5-13-2092, and 5-13-2093. In addition, the following components shall be furnished with each kit in the quantities indicated, for use in the erection, installation and blocking of the shelving in each MILVAN.

a. Two pounds of box nails conforming to 3.2.3.1
b. Three pounds of common nails conformin to 3.2.3.2
c. One piece of 2" x 4" x 90" lumber conforming to 3.2.2
d. Four pieces of 2" x 4" x 78-1/2" lumber conforming to 3.2.2
e. Twelve pieces of 2" x 4" x 93-1/2" lumber conforming to 3.2.2
f. One piece of 1/2" x 17-3/4" x 81-1/2" plywood conforming to 3.2.1
g. Two pieces of 1/2" x 11-3/4" x 81-1/2" plywood conforming to 3.2.1
h. Sixteen pieces of 1/2" x 40" x 93-1/2" plywood conforming to 3.2.1
i. Two pieces of 1/2" x 40" x 90" plywood conforming to 3.2.1

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3.4 Dimensional tolerances. The component parts of the shelving shall be of such dimensions so that when the shelving is assembled the overall height, width and depth dimensions shall be within plus or minus 1/8 inch of the overall dimensions shown on the drawings. Where dadoes are required, the depth of each dado shall be as shown on the drawing. In no case shall the material thickness between opposing dadoes on the same divider be less than 1/4 inch. All saw cuts and dadoes shall be square with the adjacent edges within ± 1 degree.

3.5 Erection instruction. One reproduced copy of each of the following drawings, shall be included with each shipping container, for the kit specified:

a. Kit I - Drawings 5-13-2414, 5-13-2415, 5-13-2416, and 5-13-2417
b. Kit II - Drawings 5-13-2418, 5-13-2419, and 5-13-2420
c. Kit III - Drawings 5-13-2086, 5-13-2087, 5-13-2088, 5-13-2089 and 5-13-2091
d. Kit IV - Drawings 5-13-2087, 5-13-2090, 5-13-2091, 5-13-2092 and 5-13-2093

3.6 Workmanship. Component parts of the shelving shall be free of splinters, and saw and dado slashes. Assembly of the kits shall not require modification or reworking of any components to conform to overall dimensional requirements and assembly configuration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, the shelving kit shall be examined for defects in table I, 4.3.2.2, and 4.3.2.3. The presence of any defects or failure to meet any of the requirements shall be cause for rejection of the first article.

4.3 Inspection. Sampling for inspection shall be made in accordance with the provisions set forth in MIL-STD-105, unless otherwise indicated hereinafter.

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4.3.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Inspection of the end item. The lot shall be all like knocked down shelving kits offered for inspection at one time. The sample unit shall be a complete kit.

4.3.2.1 Visual examination of knocked down shelving kits. Knocked down shelving kits shall be examined for defects in table I. The inspection level shall be S-4 with an acceptable quality level (AQL) of 6.5, expressed in terms of defects per hundred units.

TABLE I. Visual examination of knocked down shelving

Examine	Defect
Materials	Not as specified.
Workmanship	Splinters. Saw cuts not square with adjacent edge within 1 degree. Dadoes not square with adjacent edge within 1 degree. Saw or dado slashes. Like components not uniform size. Dadoes not equally spaced end to end. Wrong components. Any components missing (including nails, blocking material and erection instructions).

4.3.2.2 Dimensional examination. The components of each kit shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units.

4.3.2.3 Examination of assembly. Sample units from the inspection lot defined in 4.3.2 shall be assembled (temporarily nailed together as required) and examined for fit of component parts, fit of shelves, or dividers (as applicable), in dadoes, and overall assembled dimensions. Any evidence of noncompliance shall be classified a defect. The inspection level shall be S-2, with an AQL of 6.5, expressed in terms of defects per hundred units. This examination need not be repeated during

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the life of the contract, providing the first inspection lot is acceptable under the inspection level and AQL noted above, and providing units in subsequent lots are identical in all respects to the units in the first inspection lot.

4.3.3 Inspection of preparation for delivery. Examination shall be made to determine compliance with the applicable packaging, packing, and marking requirements specified in section 5. The sample unit for this examination shall be one complete kit fully prepared for delivery, except that the container need not be closed. Examination for closure defects shall be made on shipping containers fully prepared for delivery. The lot shall be all like kits offered for inspection at one time. Defects shall be as indicated in table II. The inspection level shall be S-2 with an AQL of 15, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Packaging	Any component missing
	Not bundled in assemblies
	Not flat bundle with smaller components inside
	Strapping not as specified
	Buckles used on nonmetallic straps
	No edge protectors under steel straps
	Erection instructions not packaged as specified
	Nails not packaged as specified
Packing	More of less than one complete kit
	Container not as specified
	Arrangement of components within container not as specified
	Erection instructions not secured as specified
	Alternate container base not constructed as specified
	Securing of alternate containers, cover and base not as specified
	Strapping not as specified
	Buckles used on nonmetallic strapping
Marking	Omitted, incorrect, not legible, of improper size, location, sequence or method of application

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A.

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5.1.1 Level A

5.1.1.1 Bin assemblies, center, A, B, C, D, E, F, G, H, I and J. Components of each bin assembly, as shown on applicable drawings shall be nested by assembly in the most compact manner into a flat bundle. The smaller components shall be placed within the interior of the bundle. Each bundle shall then be secured from movement with minimum 3/8 x 0.015 inch strapping conforming to type II or III of PPP-S-760 or type I, class A of QQ-S-781 or minimum 1/2 inch wide tape conforming to type III or IV of PPP-T-97. Buckles shall not be used on nonmetallic strap and steel strapping shall be provided with commercial edge protectors between the strapping and the edges of the bundle.

5.1.1.2 Erection instructions. Erection instructions for each kit (see 3.5) shall be packaged in accordance with method 1C-1 of MIL-P-116.

5.1.1.3 Nails. Box nails shall be packaged in a commercial paperboard box in two lb. quantities. Common nails shall be packed in a commercial paperboard box in one lb. and three lb. quantity in accordance with the applicable kit requirements. Each paperboard box shall be placed in a loose fitting 0.003 inch thick commercial polyethylene bag and the bag sealed.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Components comprising one complete kit, (kit I, II, III, or IV), with bin assembly components packaged as specified in 5.1 and unpackaged components, shall be packed in a compact manner in a snug-fitting shipping container conforming to type II, style A of PPP-B-650, except that the weight limitation shall be waived. Blocking members including plywood and nails shall be secured in an unused portion of the crate. The contents shall be arranged so there is at least one piece of the largest piece of plywood covering the top and bottom of the packed components within the crate. The packaged erection instructions shall be secured in a prominent place in the crate. Anchoring closure and strapping as required shall be in accordance with the appendix of PPP-C-650.

5.2.2 Level B. Components comprising one complete kit (kit I, II, III or IV) with bin assembly components packaged as specified in 5.1 and unpackaged components, shall be packed as specified in 5.2.1. Alternately the shipping container shall be in accordance with style G, class 2 of PPP-B-640 secured to a wooden base. The base dimension shall be the same as the outside length and width of the box and made from not less than three longitudinal 1 x 6 inch lumber members secured to not less than three cross 2 x 4 inch skids, two placed 1/6 the distance in from each end the third centered. Each cross member shall be secured with a minimum of three 8d commercial common nails passing through the 1 x 6 inch member

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and into the skid and clinched. The box shall be secured to the base with flathead commercial roofing nails and flat metal washers. Each box, cover, and base shall be reinforced and secured with 3 girthwise straps placed adjacent to each skid. The straps shall be minimum 3/4" x .023" in accordance with type I, class A of QQ-S-781 or minimum 850 lb. breaking strength strapping conforming to type II or III of PPP-S-760. Buckles shall not be used on non-metallic strapping.

5.2.3 Level C (commercial pack). Shelving, of one kit only, packaged as specified in 5.1 with unpackaged components, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The pack shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking. The package containing erection instructions shall be so marked. Each nested bundle of shelving shall be marked with a tag designating the applicable bin assembly and kit.

6. NOTES

6.1 Intended use. The type I shelving covered by this specification is intended for installation in type II CONEX units conforming to MIL-H-11886. The type II shelving is intended for installation in MILVAN units conforming to MIL-S-62076. Both types are used for storing and shipping repair parts, and other items of supply, to overseas users to meet the following requirements:

a. To enable troop units to have all stored items ready for immediate issue from the containers.
b. For speed and ease in locating repair parts.
c. To utilize CONEX and MILVAN units as storage warehouses.
d. To permit issue of items directly from CONEX and MILVAN units.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Kit required (see 1.2).
(c) When a preproduction sample is required (see 3.1).
(d) Selections of applicable level of packing (see 5.2).

6.3 Supersession data. This specification includes the requirements of Interim Purchase Description IP/DES S-8-6 dated 28 January 1966.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific items described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspections and approval of the first article.

Custodian:

Army - GL

Review activity:

Army - SM

Preparing activity:

Army - GL

Project No. 7125-A092

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-43894(GL)

SHELVING KITS, STORAGE AND DISPLAY, WOOD: FOR BOXES, SHIPPING, METAL (CONEX AND MILVAN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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